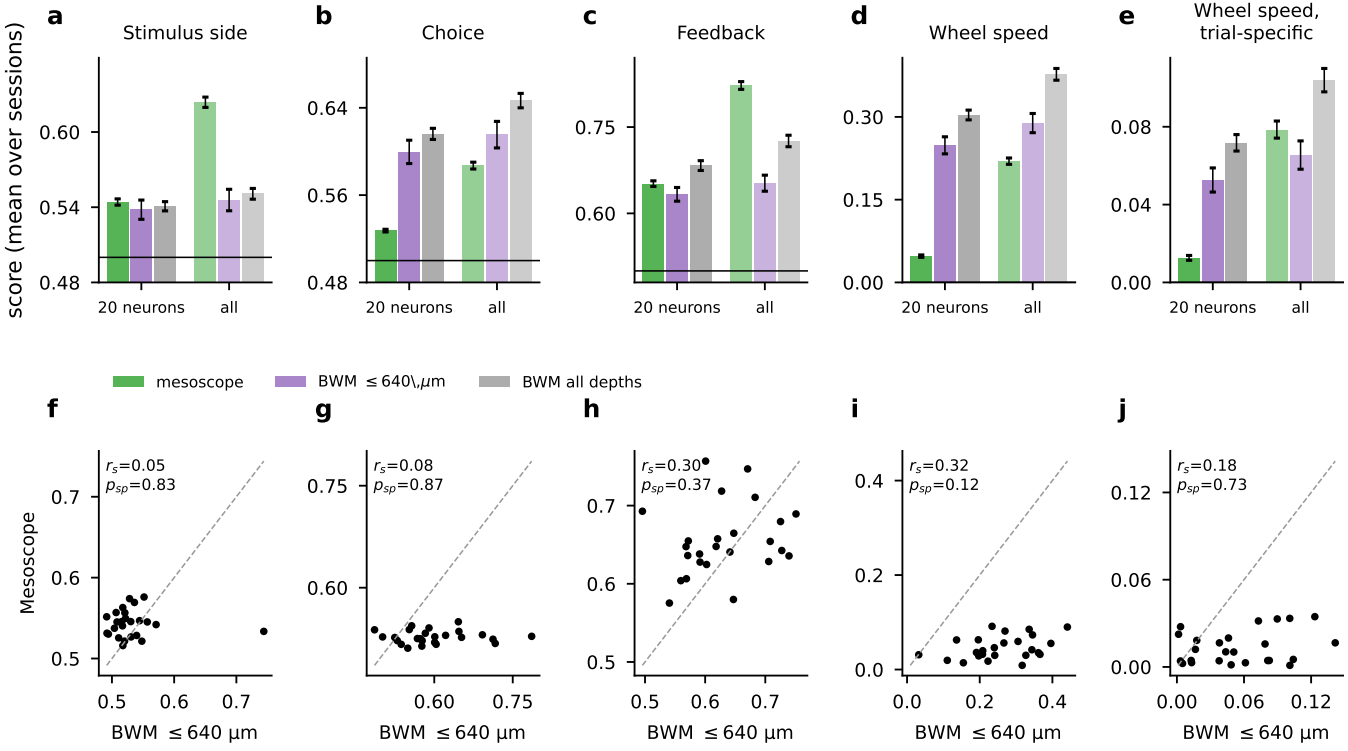

Figure 1: Summary of the revised decoding. **a–e** Mean over sessions (± s.e.m.) of each session's mean score over its 1 mm bins, from 20 random neurons (left, solid) and all neurons (right, light). **f–j** Per 1 mm bin, 20 neurons: BWM $\leq$640 μm (x) vs mesoscope (y), the 25 bins covered by both; Spearman r_s, naive p, and p_{sp} against 1000 variogram-matched surrogates of either map (larger p reported). Decoding per session and bin (sessions with $\geq$50 trials and $\geq$20 neurons in the bin), 5-fold cross-validation over trials. Binary variables: L2 logistic regression, balanced accuracy (chance 0.5). Wheel speed in six 0.2 s bins from -0.2 to 1 s around movement onset: ridge regression on the activity of the previous, same and next bin (penalty by exact leave-one-trial-out error), cross-validated R^2; trial-specific: the same with the mean speed profile of the training trials subtracted, so only trial-to-trial differences in speed count. **a** Stimulus side, 20 neurons: mesoscope 0.544, BWM $\leq$640 μm 0.538, BWM all depths 0.541; all neurons: mesoscope 0.624, BWM $\leq$640 μm 0.546, BWM all depths 0.551. **b** Choice, 20 neurons: mesoscope 0.527, BWM $\leq$640 μm 0.600, BWM all depths 0.616; all neurons: mesoscope 0.587, BWM $\leq$640 μm 0.615, BWM all depths 0.647. **c** Feedback, 20 neurons: mesoscope 0.652, BWM $\leq$640 μm 0.633, BWM all depths 0.683; all neurons: mesoscope 0.822, BWM $\leq$640 μm 0.652, BWM all depths 0.726. **d** Wheel speed, 20 neurons: mesoscope 0.048, BWM $\leq$640 μm 0.248, BWM all depths 0.303; all neurons: mesoscope 0.220, BWM $\leq$640 μm 0.289, BWM all depths 0.377. **e** Wheel speed, trial-specific, 20 neurons: mesoscope 0.013, BWM $\leq$640 μm 0.053, BWM all depths 0.072; all neurons: mesoscope 0.079, BWM $\leq$640 μm 0.065, BWM all depths 0.104. **f** Stimulus side: r_s = 0.05, naive p = 0.80, p_{sp} = 0.83. **g** Choice: r_s = 0.08, naive p = 0.69, p_{sp} = 0.87. **h** Feedback: r_s = 0.30, naive p = 0.15, p_{sp} = 0.37. **i** Wheel speed: r_s = 0.32, naive p = 0.11, p_{sp} = 0.12. **j** Wheel speed, trial-specific: r_s = 0.18, naive p = 0.38, p_{sp} = 0.73.